

BT6270| BE22B048

Report: Assignment 1

Hodgkin Huxley Neuron Model



Files In the Zip Folder

1. Problem Statement, BT6270_Assignment-1.docx.
 2. IPython notebook, *be22b048_as1_HH_dynamics.ipynb*: contains the complete code for the model and different simulation conditions.
 3. CSV file, *hh_firing_behavior_I0_I160_sweep*: A CSV file that contains the number of action potentials and firing rate of the model for different constant input currents.
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Analysis of Hodgkin-Huxley Neuron Dynamics

This report details the simulation of a Hodgkin-Huxley (HH) neuron model to explore its firing patterns in response to varying levels of constant applied current.

The analysis involves two main stages:

- A classification of the model dynamics viz., no Action Potentials (APs), finite APs, continuous APs and AP distortion across a wide range of currents
- A detailed examination of representative examples from each identified regime.

Model Parameters

I used the Hodgkin-Huxley model, with parameter values derived from their Nobel Prize-winning experiments on the giant squid axon.

The key parameters are:

- **Maximal Conductances:**
 - $g_{Na_{max}}$: **120.0 mS/cm²** (Sodium)
 - $g_{K_{max}}$: **36.0 mS/cm²** (Potassium)
 - g_l : **0.3 mS/cm²** (Leak)
- **Reversal Potentials:**
 - E_{Na} : **50 mV** (Sodium)
 - E_K : **-77 mV** (Potassium)
 - E_l : **-54.387 mV** (Leak)
- **Membrane Capacitance:**
 - C_m : **1.0 μ F/cm²**

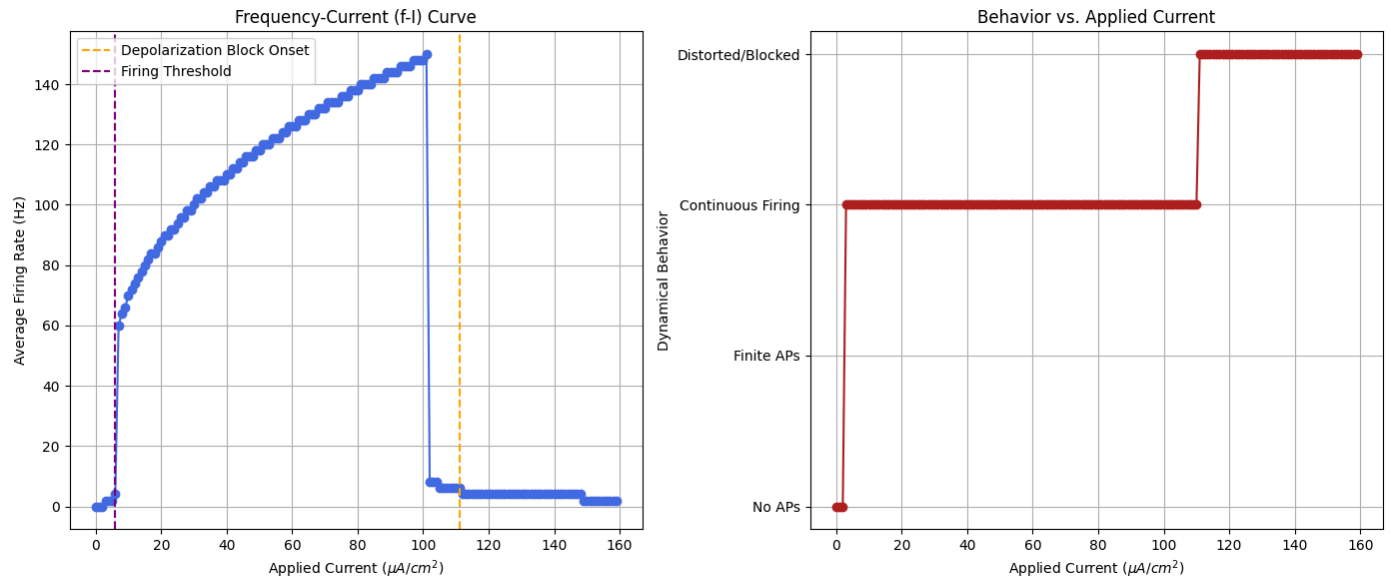
Choosing these values accurately reproduces the experimentally observed shape and propagation of an AP.

Simulation and Analysis Assumptions

I made the following assumptions to facilitate the analysis:

1. Numerical Integration: The model's differential equations were solved using the Forward Euler method, with a time step (Δt) of 0.01 ms. The step size is a compromise between numerical stability and computational cost.
2. Simulation Duration: A total simulation time (T) of 0.5 second (500 ms) was used for the current sweep. This duration is long enough to accurately calculate the steady-state firing rate and to observe whether the neuron's firing adapts or ceases over time.
3. Initial Conditions: The simulation starts from the resting state of the neuron ($v = -65$ mV), with the gating variables (m, h, n) set to their corresponding steady-state values at rest. This ensures that the neuron's response is due to the applied stimulus and not a transient effect of arbitrary initial conditions.

Classification of Firing Patterns



Below, I describe the tools and logic used to classify the voltage trace into the different regimes.

Imported Functions:

- `scipy.signal.find_peaks`: This function was used to detect action potentials (APs) as it identifies local maxima in the voltage trace.
- `numpy.mean`: Used to calculate the average membrane potential during the final phase of the simulation, which I compared with a threshold for detecting depolarization block.

Classification Logic and Assumptions:

Dynamic Classification Via `analyze_voltage_trace_app1`

The resulting voltage trace for each input current value was passed to the `analyze_voltage_trace_app1` to classify the firing based on its features.

Imported Functions:

1. `scipy.signal.find_peaks`: Used to detect action potentials (APs) by identifying local maxima in the voltage trace.
2. `numpy.mean`: Used to calculate the average membrane potential during the final phase of the simulation.

Dynamic Classification Logic

1. Spike Detection: An AP was counted if its peak potential exceeded -20 mV, as specified by the `height=-20` argument in `find_peaks`.
2. Depolarization Block Check: The neuron was classified as "Distorted/Blocked" if the mean voltage in the last 20% of the simulation was greater than -40 mV. This condition takes precedence and identifies the blocked state regardless of initial spike activity.
3. Finite vs. Continuous Firing: If not blocked, the behavior was determined by the timing of the last spike. If it occurred before 80% of the simulation time had elapsed, the regime was "Finite APs"; otherwise, it was "Continuous Firing."

Current Ranges for Dynamical Regimes

Behaviour	Range of Applied Current I_{app} ($\mu\text{A}/\text{cm}^2$)
No Action Potentials	$\sim 0 - 5$
Finite APs	$\sim 7 - 9$
Continuous Firing	$\sim 10 - 113$
Depolarisation Block	$\sim \text{Greater than } 113$

Step 2: Hardcoding Current Values For Classification Via `analyze_voltage_trace_app1`

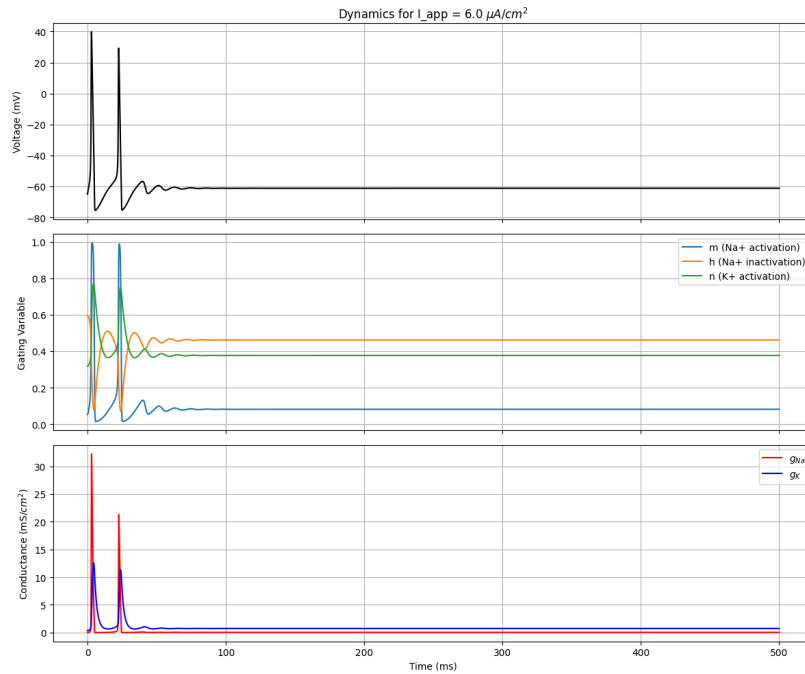
I observed that the frequency of firing (along with other properties) monotonically increased or decreased in a given regime. Thus, after the Dynamic Classification was complete, and the initial "Behavior vs. Applied Current" and "Firing Rate vs Applied current" plot was generated, specific **hardcoded current values** were chosen to serve as clear boundaries to separate the regimes. I mostly used visual inspection of voltage vs time plots to determine the exact boundary values.

Selection of I_{app} and Plotting for Each Regime

After the complete classification process, values belonging to each regime were chosen to serve as a representative example.

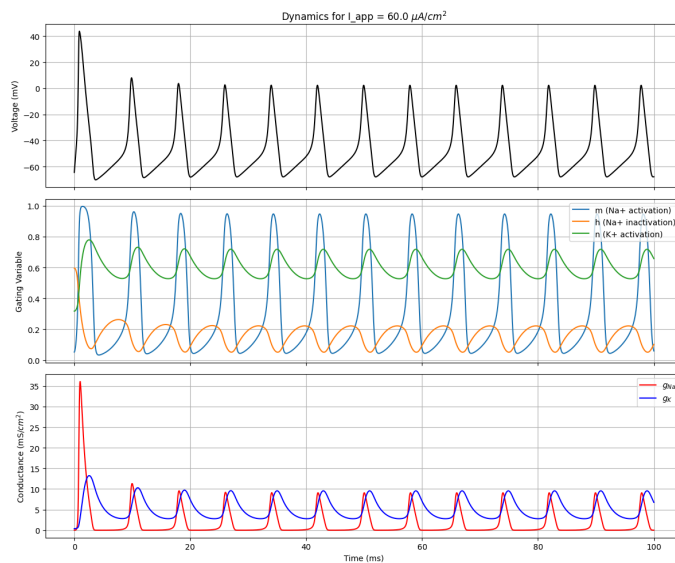
The selected representative currents in the notebook are:

- $I_{no_ap} = 3.0 \mu\text{A}/\text{cm}^2$
- $I_{finite_aps} = 7.0 \mu\text{A}/\text{cm}^2$
- $I_{continuous} = 60.0 \mu\text{A}/\text{cm}^2$
- $I_{distorted} = 140.0 \mu\text{A}/\text{cm}^2$.



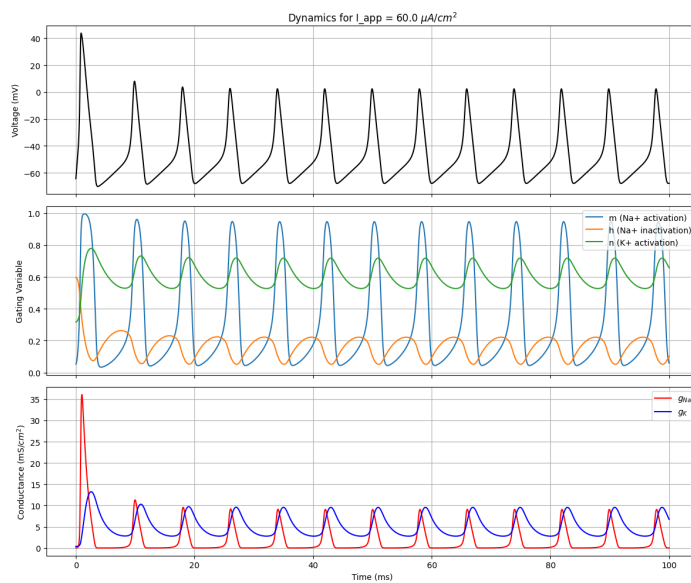
Case 1: No Action Potentials

Observation: The current causes a subthreshold depolarization insufficient to trigger a spike. The voltage-gated sodium channels do not activate sufficiently



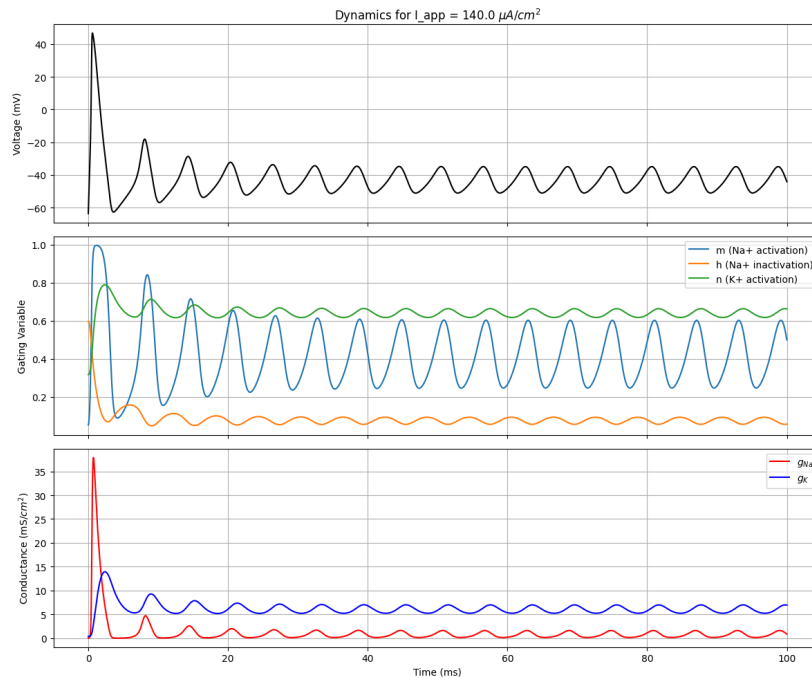
Case 2: Finite Action Potentials

Observation: The neuron fires a short burst and then stops, a behavior known as adaptation. This is driven by the slow accumulation of the potassium current (I_K), which eventually becomes strong enough to prevent the membrane from reaching threshold again.



Case 3: Continuous Firing

- **Observation:** The current reliably overcomes the potassium current, leading to a rhythmic, continuous train of action potentials



Case 4: Distorted / Depolarization Block

Observation: The neuron fires once (or not at all) and then the membrane potential gets stuck at a highly depolarized level. The extreme input current prevents the sodium channel inactivation gate (h) from recovering, shutting down the spike-generating mechanism.